

Topology optimization in fluid mechanics using continuous adjoint and the cut-cell method

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ARTICLE INFO

Keywords:

Cut-cell method
Topology optimization
Continuous adjoint method
Laminar incompressible flow

ABSTRACT

A topology optimization method for steady-state flows of incompressible fluids which is capable of imposing accurate boundary conditions along the solid walls of the sought fluid paths is presented. In each topology optimization cycle, body-conforming Cartesian meshes are generated around shapes of any complexity by tracing the fluid-solid interfaces and a cut-cell flow solver is implemented together with its adjoint. Sensitivity derivatives are computed along the fluid-solid interfaces via the continuous adjoint method. Changes in topology are caused by expressing the computed sensitivity derivatives w.r.t. an auxiliary background material distribution, that helps updating the fluid-solid interfaces. The proposed method performance is assessed on three 2D benchmark examples and a 3D case. Two out of the three 2D examples are also solved using a porosity-based topology optimization approach in which impermeable regions are penalized by a Brinkman term and useful conclusions are drawn. For a fair comparison, designs optimized using the porosity-based method are re-evaluated after extracting fluid-solid interfaces from the computed porosity fields.

1. Introduction

Topology Optimization (TopO) [1] involves the pursuit of obtaining a better qualified design for a specific application by altering the material composition within its design space, while also satisfying the governing equations and design constraints. Introduced and almost matured in solid mechanics [2], TopO has been extended to various scientific fields due to the flexibility it offers, as it is not bounded by an a priori defined shape or topology. In fluid mechanics in specific, TopO has successfully been applied in Stokes [3], laminar [4,5], and turbulent [6,7], steady, and unsteady [8–10] flows. TopO's predominant methodology in fluid mechanics is via the porosity-based approach (a.k.a. the Brinkman penalization method), in which a porous material is introduced and controlled, giving rise to the design variables of the case. Solid regions are characterized by low porosity or high impermeability, impeding fluid flow by enforcing a zero velocity. The ability to radically change the topology of the solution during the optimization cycles and the fairly simple implementation are the main advantages of the porosity-based TopO. Its main weakness is related to accuracy issues, as it cannot account for the accurate effect of solid walls on the computed flow field. Also, stair-case effects, instead of curved boundaries, may result, whereas narrow solid regions cannot accurately be

simulated, as pressure diffuses through the solid regions [11,12]. This is why, optimized solutions computed by standard porosity-based TopO require re-evaluation using flow solvers running on body-fitted meshes. The extraction of the optimal body shape from the optimized porosity field becomes necessary and this is not a straightforward task which, among other, may impair the quality of the optimized solution since the re-evaluation usually computes a (slightly) different performance value. For this reason, after parameterizing the so-extracted boundaries, a shape optimization (ShpO) using this parameterization may follow to further refine the optimized solution [13].

These shortcomings of the porosity-based approach, the increasing complexity of TopO applications as well as the need to compute sharp fluid-solid interfaces [12,14,15], have rejuvenated surface-capturing approaches, in which solid boundaries are reconstructed during the optimization process rather than upon its completion, usually via level-set computations on fixed background meshes [11,16,17]. Capturing the solid surface anew within each optimization cycle allows for the imposition of accurate boundary conditions. In level-set TopO methods, the zero-level iso-surfaces are displaced by the computed sensitivity derivatives; these support topological changes such as boundary-merging, boundary-splitting, and boundary disappearance [18]. However, for the generation of new solid boundaries, i.e. the appearance of new fluid-

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<https://doi.org/10.1016/j.camwa.2021.06.002>

Received 13 December 2020; Received in revised form 26 March 2021; Accepted 3 June 2021

Available online 18 June 2021

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solid interfaces inside the fluid domain, extra treatment is required [19].

The XFEM approach [9,11,20], is an alternative way of mitigating the shortcomings of the porosity approach. In XFEM, the TopO algorithm combines a level-set description of the fluid-solid interface and an extended finite element formulation of the flow equations. It may capture discontinuities in the solution of equations within an element intersected by the solid by enriching the shape functions and enforcing the interface conditions along the intersection with accuracy. Similarly to the herein proposed method, the governing equations are only solved in the flow domain, so spurious pressure diffusion through the solid is avoided and the computational cost is reduced.

In this paper, the cut-cell method [21] is used within a TopO algorithm to generate Cartesian meshes with cells intersected by the solid walls to be designed and impose accurate boundary conditions along them [22], overcoming the need of solving the Hamilton-Jacobi equation, as in most level-set methods. The cut-cell method belongs to the general class of immersed boundary methods [23], in which the Cartesian cells crossed by the solid boundaries are adapted to the case by discarding their solid parts, giving rise to body-conforming cut cells. Within each optimization cycle, crisp fluid-solid interfaces are created to apply exact boundary conditions. Since a common problem in cut-cell methods is the treatment of very small cut-cells, a cell-merging algorithm [24] is additionally implemented to alleviate stability issues.

In gradient-based TopO, extrema can be found using the gradient of the objective function computed via the adjoint method. Dealing with a large number of design variables, the cost of obtaining the gradient of a function is approximately equal to the cost of numerically solving the governing equations. The adjoint problem emerges by differentiating either the continuous or the discrete form [25] of the governing PDEs. Continuous and discrete adjoints have their own advantages, [26–28], but such a discussion is beyond the scope of this paper. Here, the continuous adjoint method is used. The geometric sensitivities of the evolving fluid-solid boundary, required to compute the said gradient, are computed analytically by differentiating the Cartesian cell cutting procedure [29].

This work presents the development of a TopO framework in fluid mechanics implementing the cut-cell method in conjunction with a background material distribution (α) which gives rise to the design variables. The fluid equations are solved by the cut-cell method, and the material variables are processed to provide the fluid-solid interface, so as to impose boundary conditions. Therefore, blockage terms and the subsequent material interpolations [30] used in standard porosity TopO are no more needed. The paper is organized as follows: Section 2 presents the objective and constraint functions this paper is dealing with. Then, in Section 3, the methodology of generating body-fitted Cartesian meshes, the flow equations, and the development of the continuous adjoint method are described. A brief note on differences from the standard porosity-based TopO method, with comparative results in a properly selected case, is also included. Finally, in Section 4, the proposed method is used to solve three 2D TopO examples, two of which include comparisons with designs obtained via the porosity-based TopO, and a 3D case.

2. Topology optimization problem definition

Herein, TopO aims at the minimization of the volume-averaged total pressure losses between fixed inlets (S_I) and outlets (S_O). This target quantifies the total mechanical energy loss due to friction forces on the solid walls and among fluid particles inside the fluid domain. The objective function is expressed as

$$J = - \int_{S_I} p_t u_n dS - \int_{S_O} p_t u_n dS \quad (1)$$

where $p_t = p + \frac{1}{2} \mathbf{u} \cdot \mathbf{u}$ is the total pressure, p the static pressure (both divided by the constant fluid density), and $u_n = \mathbf{u} \cdot \mathbf{n}$ with $\mathbf{u}$ representing

the fluid velocity vector and $\mathbf{n}$ the outward unit vector normal to the I/O boundaries. Signs in Eq. (1) are determined by the outward (w.r.t. the fluid domain) direction of $\mathbf{n}$. At the same time, the flow equations $\mathbf{R} = \mathbf{0}$ and an inequality geometric constraint g_v should be satisfied. g_v is related to the volume in 3D or the area in 2D occupied by the fluid (Ω_f)

$$g_v = \frac{\int_{\Omega_f} d\Omega_f}{\int_{\Omega} d\Omega} - V_{tar} \leq 0 \quad (2)$$

which should not exceed a user-defined volume ratio (V_{tar}) of the background domain Ω . The design variables, namely the values of the material field on the nodes of the background mesh, are bounded by $0 \leq \alpha^m \leq 1$, $\forall m = 1, \dots, M$, where low and high α values indicate fluid and solid nodes respectively.

3. The cut-cell TopO algorithm and its steps

The material field α is mapped onto boundary indicator variables ϕ in order to trace the fluid-solid interfaces, compute their intersections with the background Cartesian mesh over Ω and, thus, create the cut-cells. Therefore, the flow and adjoint equations are solved on the cut-cell mesh, with clear solid walls. In contrast to porosity-based TopO, flow and adjoint, equations are satisfied only in the fluid domain Ω_f and the adjoint problem is formulated by differentiating J w.r.t. the coordinates of points on the solid boundaries of cut-cells. The adjoint solution gives a sensitivity map and α is updated accordingly. The background Cartesian mesh is used for both storing α and creating the cut-cells for the solution of the flow and adjoint equations. All constituents of the proposed TopO method are analyzed, one-by-one, below.

3.1. Generation of a body-fitted Cartesian mesh using the cut-cell method

Considering the field of nodal α values (see following Section), the solid and fluid regions and their interface are identified using the nodal boundary indicator variables ϕ . Thus, the body-fitted Cartesian mesh formed by cut-cells and Cartesian cells entirely residing in the fluid domain, is generated.

3.1.1. Computation of the boundary indicator field

The nodal material field α is mapped onto the boundary indicator variables ϕ using a two-step approach. Compared to standard level-set approaches [31], this is a much simpler approach as it avoids the solution of the Hamilton-Jacobi equation to displace the interface. In the first step, α is mapped onto a smoothed intermediate field, $\tilde{\alpha}$, [32],

$$\tilde{\alpha}^m = \frac{\sum_n \alpha^n w^n}{\sum_n w^n}, \quad w^n = r - \|\mathbf{x}^n - \mathbf{x}^m\|_2 \quad (3)$$

where the summation is performed over all nodes lying within a circle/sphere of radius $r = d^r h$ centered at node m , where d^r is a user-defined multiplier and h is the length of a Cartesian cell edge. The purpose of this step is to accelerate the convergence of the optimization by increasing the zone of influence of the design variables, as well as to avoid crude solid boundaries [18].

In the second step, the smoothed field $\tilde{\alpha}$ is steepened using an approximate Heaviside function, [32],

$$\phi^m = \frac{1}{2} - \frac{\tanh(0.5\beta) + \tanh(\beta(\tilde{\alpha}^m - 0.5))}{2 \tanh(0.5\beta)} \quad (4)$$

centered at $\tilde{\alpha}^m = 0.5$, and mapped onto the boundary indicator field ϕ , such that $\phi^m > 0$ refers to fluid, $\phi^m < 0$ to solid, and $\phi^m = 0$ to the fluid-solid interface. Eq. (4) makes use of the steepening parameter β . The performed studies showed that $\beta = 12$ and $3 \leq d^r \leq 6$ lead to smooth boundaries and facilitate the convergence of the optimization loop [33, 34].

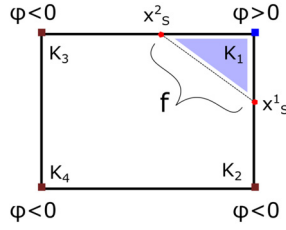


Fig. 1. Cartesian cell with ϕ values on its nodes, (K_1 , K_2 , K_3 , K_4). The opposite signs of ϕ at K_1 , K_2 , and K_1 , K_3 lead to the intersection points x_S^1 and x_S^2 . The resulting cut-cell is a triangle with its vertices at K_1 , x_S^1 and x_S^2 .

3.1.2. Generation of cut-cells based on ϕ

Having obtained the nodal ϕ variables, intersection points of the zero- ϕ contour can be computed along edges of the Cartesian cell, the nodes of which have ϕ values with opposite signs. Cut-cells are defined by computing these intersection points and connecting them with the Cartesian cell nodes that have $\phi^m > 0$. The necessary geometric quantities of cut-cells, such as cell volumes, surface areas and unit vectors normal to faces, can be computed based on the points that surround the fluid part inside the Cartesian cell. The intersection points x_S and their derivatives w.r.t. ϕ are computed as

$$x_S = x^A - \frac{x^B - x^A}{\phi^B - \phi^A} \phi^A \quad (5)$$

$$\frac{\partial x_S}{\partial \phi^i} = \frac{x^B - x^A}{(\phi^B - \phi^A)^2} \left(\phi^A \frac{\partial \phi^B}{\partial \phi^i} - \phi^B \frac{\partial \phi^A}{\partial \phi^i} \right), \quad i = A, B \quad (6)$$

where A and B are two mesh nodes connected by an edge with $\phi^A \phi^B \leq 0$. In Fig. 1, an illustrative 2D example is given where a triangular cut-cell is created based on the ϕ values of the Cartesian cell nodes.

In 3D, Cartesian cells having more than 3 intersection points along their edges occur frequently. To avoid the creation of faces formed by points that are not co-planar on the solid boundary and could compromise the solution accuracy, additional points are introduced at the average of the intersection points inside the Cartesian cell. The intersection points along the cell edges are then connected with the additional point to form triangles on the solid boundary, Fig. 2b. In case a Cartesian cell is intersected twice by the solid boundary, double-cut cells are formed, Fig. 2c; for them, the intersection points split into two clusters and an additional point per cluster is introduced.

In practice, the resulting cut-cells can be very small and this may negatively impact stability. This is why an algorithm [24] is used to merge cut-cells with a volume less than a threshold value (5% of the Cartesian cell volume) with a bigger neighboring cell, Fig. 3. Connectivity information and geometric quantities are passed on to the merged cell replacing the constituent cells.

3.2. Flow equations

The Navier-Stokes equations for the laminar flow of an incompressible fluid are solved in Ω_f , as defined by tracing the boundaries of the fluid based on the available α field in each optimization cycle. The governing equations in vector form read

$$\Gamma \frac{\partial \mathbf{V}}{\partial \tau} + \underbrace{\frac{\partial \mathbf{F}_j}{\partial x_j}}_{\mathbf{R}} = \mathbf{0} \quad (7)$$

with,

$$\Gamma = \begin{bmatrix} \frac{1}{\beta^2} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{F}_j = \mathbf{F}_j^{inv} - \mathbf{F}_j^{vis} = \begin{bmatrix} u_j \\ u_1 u_j + \delta_j^1 p \\ u_2 u_j + \delta_j^2 p \\ u_3 u_j + \delta_j^3 p \end{bmatrix} - \begin{bmatrix} 0 \\ \tau_{1j} \\ \tau_{2j} \\ \tau_{3j} \end{bmatrix} \quad (8)$$

where Γ is the artificial compressibility preconditioner [35], $\mathbf{V} = [p \mathbf{u}]^T$ is the flow variables' vector and $\mathbf{F}_j$ stands for the flux vector with its inviscid and viscous parts. β is the artificial compressibility parameter, τ the pseudo-time, δ_j^i the Kronecker delta, $\tau_{ij} = \nu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$ the shear stresses, and ν the kinematic viscosity.

In the cases of this paper, a uniform velocity at S_I , a uniform pressure at S_O , and the no-slip condition along the solid walls S_W are imposed.

3.3. Formulation of the continuous adjoint equations and sensitivity derivatives

The adjoint method computes the gradient of J w.r.t. the design variables α . The adjoint equations are derived by augmenting Eq. (1) with the field integral of all flow equations residuals multiplied by adjoint fields ψ , thus forming the Lagrangian $L := J + \int_{\Omega_f} \psi^T \mathbf{R} d\Omega_f$. L is differentiated w.r.t. the coordinates of the points on the solid boundaries x_S^n , $n = 1, \dots, \mathcal{N}$, with $\mathcal{N}$ being the number of solid wall points computed along the Cartesian cells edges. The mathematical formulation leads to the adjoint field equations and boundary conditions, as well as the objective function derivatives w.r.t. x_S^n . Since x_S^n depend solely on the boundary indicator field ϕ , the chain rule gives the derivatives w.r.t. ϕ and then those w.r.t. α . Therefore, in contrast to porosity-based TopO [6], the adjoint delivers a sensitivity map on the reconstructed, by the cut-cell method, solid surfaces.

Differentiating L w.r.t. x_S^n yields [36]

$$\begin{aligned} \frac{\delta L}{\delta x_S^n} &= \frac{\delta J}{\delta x_S^n} + \int_{\Omega_f} \left(\frac{\partial \mathbf{V}}{\partial x_S^n} \right)^T \left(-\mathbf{A}_j^T \frac{\partial \psi}{\partial x_j} - \frac{\partial \mathbf{F}_j^{vis\psi}}{\partial x_j} \right) d\Omega_f \\ &+ \int_S \left(\psi^T \frac{\partial \mathbf{F}_j^{inv}}{\partial x_S^n} + \tau_{ij}^\psi \frac{\partial u_i}{\partial x_S^n} - \psi_{i+1} \frac{\partial \tau_{ij}}{\partial x_S^n} \right) n_j dS \end{aligned} \quad (9)$$

where $\mathbf{A}_j = \frac{\partial \mathbf{F}_j^{inv}}{\partial \mathbf{V}}$ is the Jacobian matrix of the inviscid flux vector, $\mathbf{F}_j^{vis\psi} = [0 \ \tau_{1j}^\psi \ \tau_{2j}^\psi \ \tau_{3j}^\psi]^T$ the adjoint viscous flux vector with $\tau_{ij}^\psi = \nu \left(\frac{\partial \psi_{i+1}}{\partial x_j} + \frac{\partial \psi_{j+1}}{\partial x_i} \right)$ being the adjoint shear stresses.

The adjoint equations are derived by eliminating dependencies of $\frac{\delta L}{\delta x_S^n}$ on $\frac{\partial \mathbf{V}}{\partial x_S^n}$ throughout Ω_f . Then, pseudo-time derivatives and the artificial compressibility preconditioner are introduced into the adjoint equations which take the form

$$\Gamma^T \frac{\partial \psi}{\partial \tau} - \mathbf{A}_j^T \frac{\partial \psi}{\partial x_j} - \frac{\partial \mathbf{F}_j^{vis\psi}}{\partial x_j} = \mathbf{0} \quad (10)$$

The adjoint boundary conditions are imposed to make $\frac{\delta L}{\delta x_S^n}$, Eq. (9), independent of the derivatives of the flow variables w.r.t. x_S^n along S_I , S_O and S_W , by also accounting for the imposed flow boundary conditions. Assuming fixed S_I and S_O boundaries, the differentiation of J , Eq. (1), w.r.t. x_S^n yields

$$\frac{\delta J}{\delta x_S^n} = \frac{\delta J}{\delta \mathbf{V}} \frac{\delta \mathbf{V}}{\delta x_S^n} = - \int_{S_I} \begin{bmatrix} u_n \\ 0 \\ 0 \\ 0 \end{bmatrix}^T \frac{\delta \mathbf{V}}{\delta x_S^n} dS - \int_{S_O} \begin{bmatrix} u_n \\ u_1 u_n + p_t n_1 \\ u_2 u_n + p_t n_2 \\ u_3 u_n + p_t n_3 \end{bmatrix}^T \frac{\delta \mathbf{V}}{\delta x_S^n} dS \quad (11)$$

Along S_I and S_W , $\frac{\delta \mathbf{u}}{\delta x_S^n} = \mathbf{0}$. Along S_O , $\frac{\delta p}{\delta x_S^n} = 0$. Starting from Eq. (9) and considering Eqs. (10) and (11), zero Dirichlet $\psi_i = 0$, $i = 2, 3(,4)$ conditions are imposed along S_W . The adjoint boundary conditions along S_I and S_O should satisfy

$$(\mathbf{A}_n \mathbf{Q})^T \psi + (1 - \kappa) \mathbf{F}_j^{vis\psi} n_j + \left(\frac{\delta J}{\delta \mathbf{V}} \right)^T = \mathbf{0} \quad (12)$$

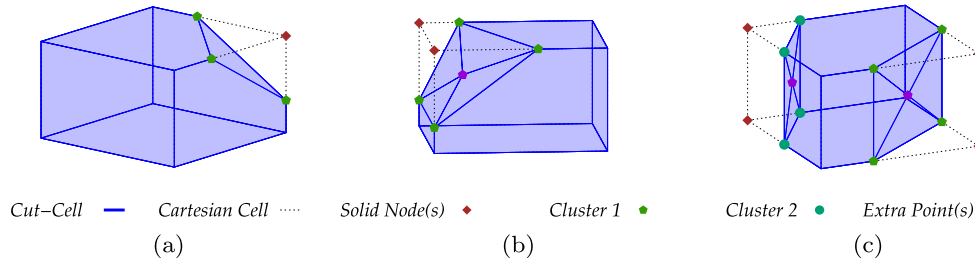


Fig. 2. Three indicative examples of 3D cut-cells where the volume encompassed by the blue lines is considered fluid. In (a), no additional point is introduced since the intersection points form a triangular solid face. In (b), having 4 non-co-planar intersection points, an additional point is introduced and 4 triangular solid faces are formed leading to a cut-cell with 10 faces in total. In (c), a double-cut cell can be seen, as intersection points split into two clusters, two additional points are introduced, and a cut-cell with 14 faces in total is created.

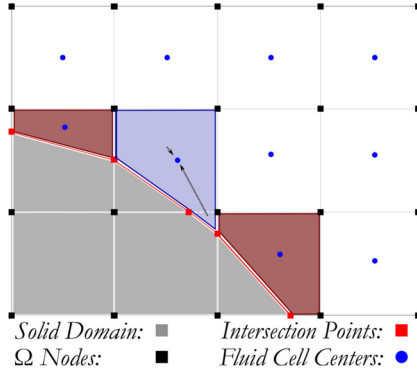


Fig. 3. Schematic of the cell-merging scheme in 2D, with part of the solid boundary, the intersection points, some non merged cut-cells (brown), and a merged cut-cell (blue). Black filled squares stand for the Cartesian mesh nodes where the α field is stored, whereas all flow and adjoint variables are stored at cell centers denoted by blue filled circles.

where

$$\mathbf{Q} = \begin{bmatrix} \kappa & 0 & 0 & 0 \\ 0 & 1-\kappa & 0 & 0 \\ 0 & 0 & 1-\kappa & 0 \\ 0 & 0 & 0 & 1-\kappa \end{bmatrix}, \quad \kappa = \begin{cases} 0, & \text{if } S_O \\ 1, & \text{if } S_I \end{cases} \quad (13)$$

and $\mathbf{A}_n = \mathbf{A}_j n_j$. By satisfying the adjoint boundary conditions, the sensitivity derivatives are initially computed as

$$\frac{\delta J}{\delta \mathbf{x}_S^n} = - \int_{S_w} \left(\psi_1 n_j + \tau_{ij}^\psi n_i \right) \frac{\partial u_j}{\partial x_k} \frac{\delta x_k}{\delta \mathbf{x}_S^n} dS \quad (14)$$

standing, practically, for a sensitivity map on the sought solid walls. Then, the computation of the gradient of J w.r.t. α requires the use of the chain rule to transform the sensitivity map computed at the solid boundary points $\mathbf{x}_S$ to derivatives at the mesh nodes of Ω , as follows

$$\frac{\delta J}{\delta \alpha^m} = \sum_l \sum_n \frac{\delta L}{\delta \mathbf{x}_S^n} \frac{\partial \mathbf{x}_S^n}{\partial \phi^l} \frac{\partial \phi^l}{\partial \alpha^l} \frac{\partial \alpha^l}{\partial \alpha^m} \quad (15)$$

where the first summation considers all nodes within the selected smoothing radius according to Eq. (3), and the second all solid wall points. Term $\frac{\partial \mathbf{x}_S^n}{\partial \phi^l}$ is given in Eq. (6), while terms $\frac{\partial \phi^l}{\partial \alpha^l}$ and $\frac{\partial \alpha^l}{\partial \alpha^m}$ are obtained by analytically differentiating Eqs. (3) and (4).

In constrained TopO, the differentiation of the constraint function is also necessary. The derivatives of the fluid volume of cut-cells w.r.t. $\mathbf{x}_S^n$ can be computed analytically, resulting to the corresponding derivatives of the constraint function

$$\frac{\delta g_v}{\delta \mathbf{x}_S^n} = \frac{1}{\int_{\Omega} d\Omega} \sum_l \frac{\delta \Omega_l^c}{\delta \mathbf{x}_S^n} \quad (16)$$

Summation is performed over the cut-cells which point n of the solid wall belongs to; $\frac{\delta \Omega_l^c}{\delta \mathbf{x}_S^n}$ is the derivative of the l^{th} cut-cell volume w.r.t. the coordinates of boundary point n . Geometric derivatives $\frac{\delta x_k}{\delta \mathbf{x}_S^n}$ and $\frac{\delta \Omega^c}{\delta \mathbf{x}_S^n}$, required by Eqs. (14) and (16), are computed as closed-form expressions, [29].

3.4. Cut-cell TopO algorithm

All constrained optimization problems were solved using the Globally Convergent Method of Moving Asymptotes (GCMMA) [37], which successively solves a number of approximate convex sub-problems, being well-suited for TopO [32]. The termination criterion was based on changes in ϕ during the last optimization cycle, namely $\|\Delta \phi\|_{\infty} < 10^{-3}$.

The proposed cut-cell TopO algorithm, which will hereafter be referred to as CCTopO, can be summarized in the following steps:

1. Initialize the nodal material field α on the fixed background mesh Ω .
2. Map α to boundary indicator variables ϕ , using Eqs. (3) and (4).
3. Compute coordinates $\mathbf{x}_S$ of solid wall points along the cell edges using Eq. (5) and create cut-cells corresponding to the body-fitted mesh in Ω_f .
4. Solve the flow and adjoint equations.
5. Compute derivatives of J and g_v w.r.t. $\mathbf{x}_S$, based on Eqs. (14) and (16).
6. Use Eq. (15) to compute derivatives w.r.t. α .
7. Employ GCMMA to update α .
8. Go to step 2 unless convergence criteria are met.

From the above algorithm, it can be seen that the boundary indicator field shares similarities with the level-set methods used in other TopO algorithms, since it implicitly describes the solid-fluid interfaces. Its main differences though are that the obtained ϕ is bounded, discontinuous, and does not satisfy the Hamilton-Jacobi equation.

3.5. Discussion - differences from standard porosity-based TopO

For the purpose of comparison, this section starts by briefly describing, the Standard Porosity TopO (SPTopO), in which the so-called Brinkman term $\mathbf{s} = \alpha_{pen} \alpha_b [0 \ u_1 \ u_2 \ u_3]^T$ is introduced into the flow equations $(\mathbf{R} := \frac{\partial \mathbf{F}_j}{\partial x_j} + \mathbf{s} = \mathbf{0})$, where α_{pen} is a penalization parameter and $\alpha_b \in [0, 1]$ the porosity at each finite volume (m). In contrast to CCTopO, the flow equations are solved over the fluid and solid domain, Ω . The design variables are the cell porosity value α_b^m ; low porosity areas correspond to solid. The derivation of the corresponding adjoint problem is well-established in the literature [6,7,13,38]. Since the Brinkman term is linear, an adjoint Brinkman term $\mathbf{s}^w = \alpha_{pen} \alpha_b [0 \ \psi_2 \ \psi_3 \ \psi_4]^T$ emerges in the adjoint equations to enforce a zero adjoint velocity in the solid regions, similarly to the flow equations. Note that, in SPTopO, the ad-

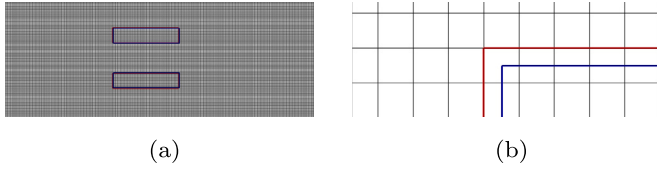


Fig. 4. Two rectilinear bodies in a channel: (a) Close-up view of the computational domain and mesh. (b) Positioning of the bodies for the first case in which the body contour coincides with mesh lines and this gives an ideal 0/1 porosity field, and the shifted contour case in which the porosity at some finite volumes takes on intermediate values.

joint solid wall conditions are imposed along the Ω boundary rather than the fluid-solid interface (i.e. the evolving Ω_f boundary). The resulting sensitivity derivatives are $\frac{\delta J}{\delta \alpha_b^m} = \alpha_{pen} u_i^m \psi_{i+1}^m \Omega^m$ corresponding to each cell, contrary to the CCTopO where the derivatives w.r.t. the design variables exist in the vicinity of the cut-cells boundaries. To impose the inequality constraint, Eq. (2), the volume/area of the fluid domain is approximated as $\Omega_f \approx \int_{\Omega} (1 - \alpha_b) d\Omega$.

The most noticeable weakness of SPTopO is its inability to accurately impose (exact) boundary conditions along the sought solid walls. Issues caused by inaccuracies in the flow solution of SPTopO approaches compared to flow solutions on body-fitted meshes, can be found in [39]. The method was used in unsteady flows and errors in the pressure fields over the solid areas of Ω significantly affected the optimization path. Also, in [11], where a U-bend duct was optimized using a SPTopO approach, it was shown that the narrow wall between the incoming and outgoing flow leads to non physical designs. In [13], the transition from TopO to ShpO was presented, illustrating differences in the volume-averaged total pressure losses before and after transition.

A comparative study of the accuracy of SPTopO flow solutions and a cut-cell solver [40] is performed. No optimization takes place in this comparison which involves only the flow solvers of the two TopO methods. The laminar flow of an incompressible fluid into a straight channel with two rectilinear internal bodies is considered. In order to investigate the SPTopO flow solver dependency upon α_{pen} and the effect of intermediate porosity values separately, two different cases are examined. In the first case, the bodies are positioned with their contours coinciding with the Cartesian mesh lines resulting to binary porosity values, whereas, in the second case, the bodies are shifted by half the size of the cell edge in both dimensions so that some cells take on intermediate porosity values, Figs. 4a and 4b.

A steady-state solution is obtained for Reynolds number $Re_H = \frac{u_m H}{\nu} = 50$, blockage ratio $r = \frac{H}{H_c} = 0.125$ and gap parameter $\gamma = \frac{d}{H} = 2.0$ and 2.04 for cases (i) and (ii), respectively. u_m is the velocity at the inlet, H_c the height of the channel, H the height of the bodies, and d the distance of the solid bodies from the closest channel wall. The outcome of four simulations using the SPTopO solver with different α_{pen} values are compared with the solution obtained by the CCTopO solver.

The computed flow fields are compared in terms of volume-averaged total pressure losses, Fig. 5. The α_{pen} value affects the obtained flow fields, as small penalization values lead to a porous body failing to enforce zero velocities, damaging the accuracy of flow predictions. On the other hand, penalization cannot be excessively high as it might cause convergence issues. Furthermore, from Fig. 5, one may conclude that, for the same α_{pen} values, the SPTopO flow solutions in cases (i) and (ii) compute quite different values for the quantity of interest, in contrast to the CCTopO flow solutions that are almost insensitive to the location of the bodies on the Cartesian mesh.

The SPTopO flow field with $\alpha_{pen} = 1000$ in both cases are further compared with the CCTopO flow field by extracting transversal pressure profiles at predefined locations. Starting with case (i), Figs. 6a and 6b show velocity magnitude iso-areas along with pressure iso-lines, and Figs. 6c and 6d the two pressure profiles at $x = -1m$ and $x = 0.8m$. At

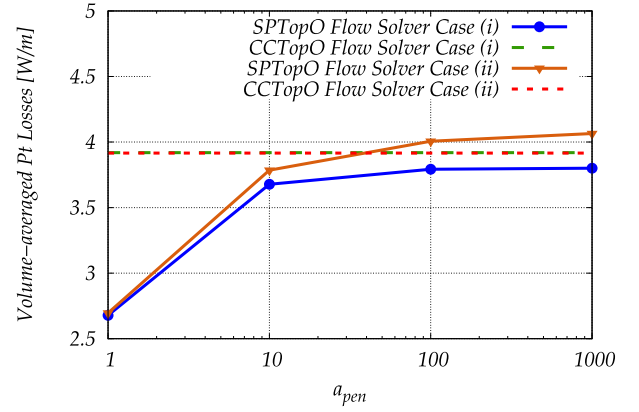


Fig. 5. Two rectilinear bodies in a channel: Comparison of the volume-averaged total pressure losses (J values) computed by the SPTopO using different α_{pen} values and the CCTopO flow solvers.

$x = -1m$, Fig. 6c, pressure profiles are comparable, despite some differences near the bodies. At $x = 0.8m$, Fig. 6d, the SPTopO flow solver consistently overestimates the pressure, compared to the CCTopO flow solver. The pressure profiles for case (ii) are shown in Figs. 7a and 7b. With the introduction of intermediate porosity values, differences from the CCTopO flow solution at $x = -1m$ become pronounced and are not restricted near the bodies anymore. At $x = 0.8m$, pressure differences are smaller, however, the pressure transversal derivatives are slightly different.

3.6. Discretization and numerical solver

A cell-centered, finite volume solver with collocated storage for all flow variables is used to solve both the flow, Eq. (7), and adjoint, Eq. (10), systems of equations [40,41]. The inviscid surface flux vectors are computed using the Roe scheme [42] and second-order accuracy is attained via the MUSCL method [43]. Flow and adjoint variable gradients, required for the MUSCL reconstruction, are computed at the cell centers using a weighted least-squares method with data from all direct neighbors. The viscous terms are computed using the gradients at neighboring cells, after applying a non-orthogonal correction. For runs on a multi-processor system, the cut-cell mesh should be partitioned. In each subdomain, the discretized PDEs are solved by means of a point-implicit block Gauss-Seidel method, and adjacent subdomains communicate via the MPI protocol.

4. Optimization results

The case studies presented illustrate the capabilities of the proposed CCTopO method and highlight its differences from SPTopO approaches due to the exact imposition of boundary conditions along the solid boundaries in each optimization cycle. For the comparison to be as clear as possible, the optimized solutions using SPTopO are re-evaluated (r-SP) by first mapping the optimized porosity field onto nodal material values and, then, to boundary indicator variables (as in CCTopO, see Section 3.1) so as to trace the solid-fluid interfaces, create cut-cells and solve the flow equations using the cut-cell solver.

4.1. 2D TopO cases

The three 2D cases considered, that target the optimal (in terms of volume-averaged total pressure losses, Eq. (1)) connection of one or two fixed inlets and one or two fixed outlets with fluid path(s) (corresponding to the fluid domain Ω_f) are shown in Fig. 8. Details related to the problems setup for each case are summarized in Table 1, where Ω_f^* and Ω_D represent the areas of the fluid and design domain respectively,

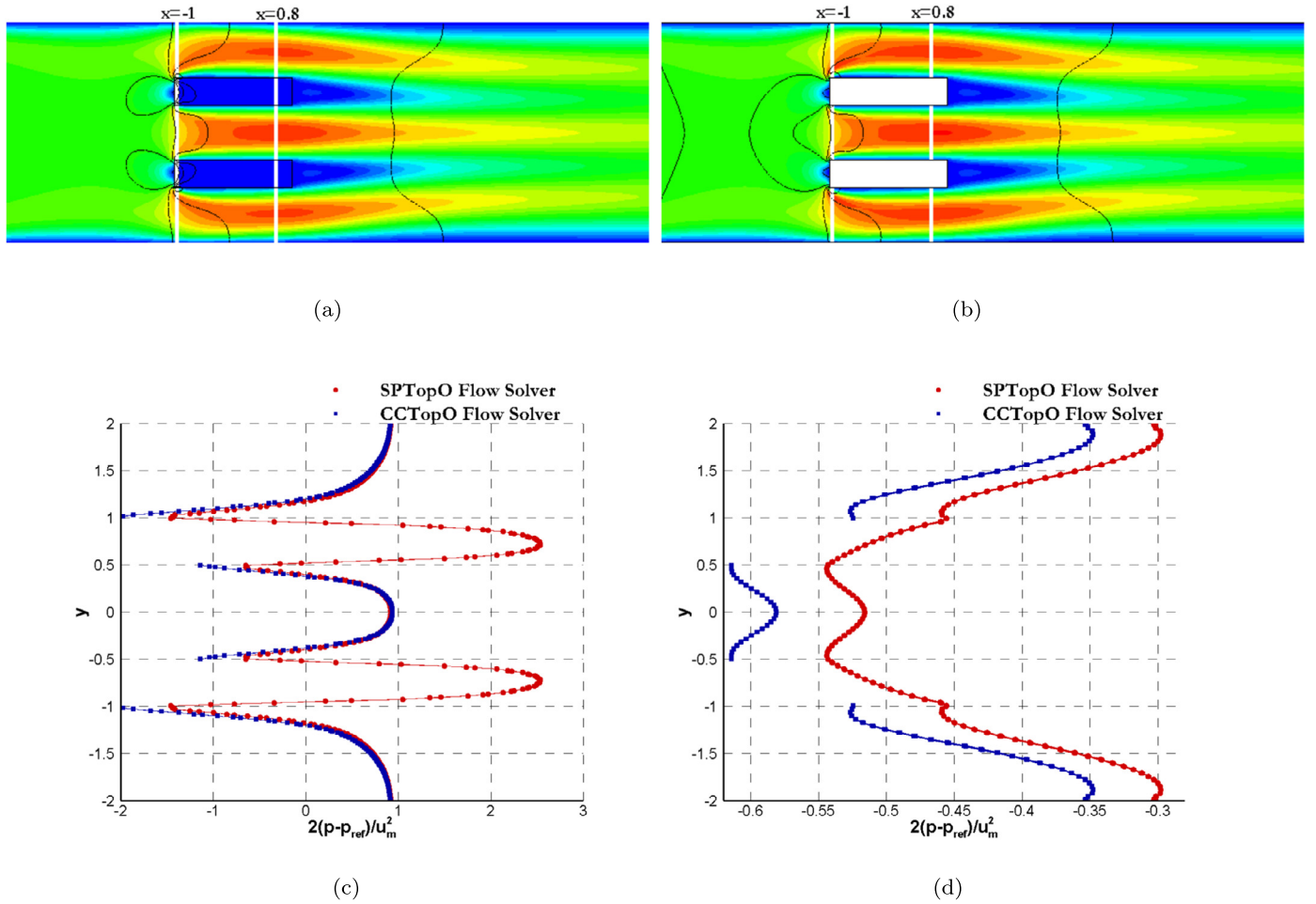


Fig. 6. Two rectilinear bodies in a channel, case (i): Velocity magnitude $\|\mathbf{u}\| \in [0, 1.9]$ iso-areas along with pressure iso-lines (a) SPTopO ($\alpha_{pen} = 1000$) and (b) CCTopO. Transversal pressure profiles at (c) $x = -1m$ and (d) $x = 0.8m$, with p_{ref} being the reference pressure defined at the outlet. Cross-sections, where pressure profiles are plotted, are shown by white lines in Figures (a) and (b).

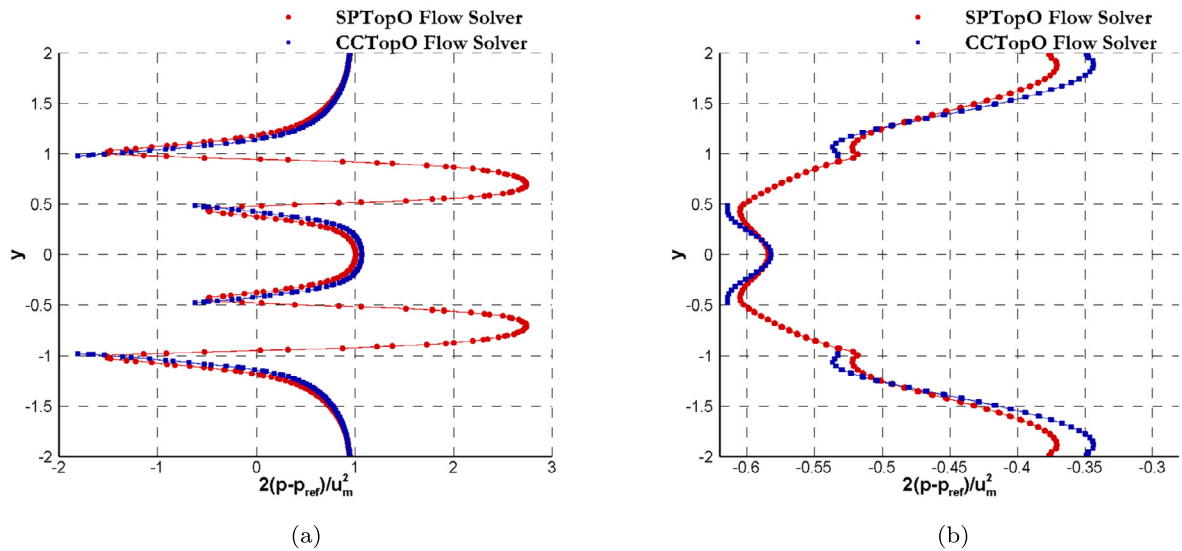


Fig. 7. Two rectilinear bodies in a channel, case (ii): Transversal pressure profiles (a) at $x = -1m$ and (b) $x = 0.8m$, with p_{ref} being the reference pressure defined at the outlet. The SPTopO solution corresponds to $\alpha_{pen} = 1000$.

and exclude the areas of the fixed inlet/outlet channels. The total number of nodes in Ω and the Cartesian cell edge length h are tabulated too. All TopO runs started by assuming that, in the initial solution, the design area Ω_D is exclusively covered by fluid.

4.1.1. Single inlet-single outlet case

TopO was performed on the same background mesh at three Reynolds numbers $Re_W = 2, 20, 200$, computed based on the inlet width W . The optimized designs are shown in Fig. 9a and agree well with

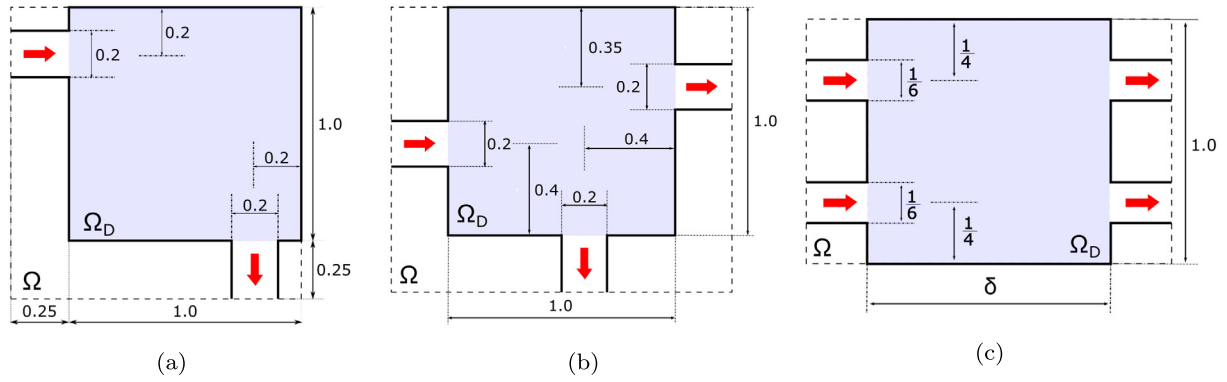


Fig. 8. 2D TopO problems description: Inlets are associated with a fixed velocity $\mathbf{u} = \mathbf{u}_{in}$, while the outlets with $p = p_{ref}$. The meshed background domain (Ω), along with the design area (highlighted area, Ω_D), i.e. the area where the design variables α are allowed to change, are shown. All dimensions are in m.

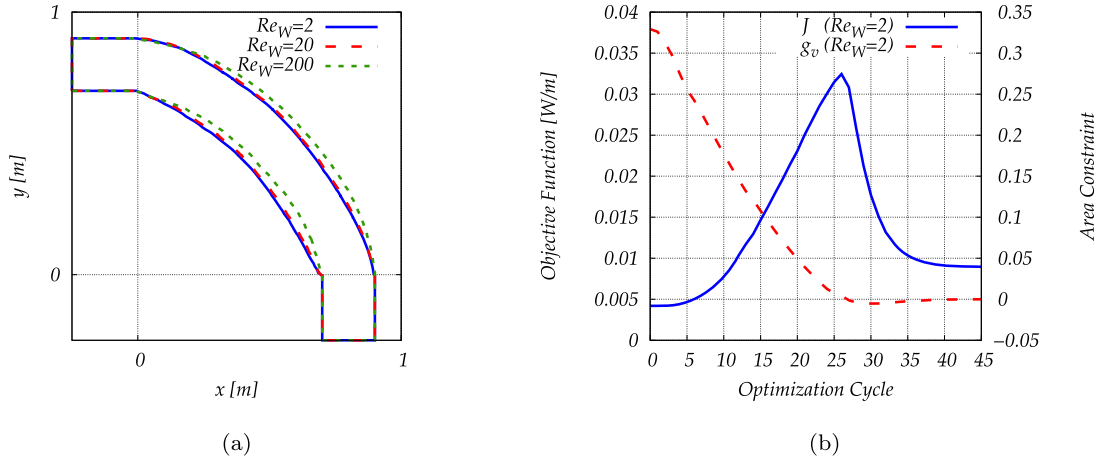


Fig. 9. Single Inlet-Single Outlet: (a) Solid boundaries of the optimized geometry (using CCTopO) for $Re_W = 2, 20, 200$. (b) Evolution of the objective and constraint functions in the $Re_W = 2$ case, with $d^r = 4$.

Table 1

2D TopO problems description: The second column describes the effective fluid volume upper bound (as given in the literature). Since in the proposed method the inequality constraint is defined based on Ω , V_{tar} is computed accordingly. In the last two columns, the total number of Ω nodes along with the Cartesian cell edge size, are given.

Case	Target $\frac{\Omega_f}{\Omega_D}$	V_{tar}	Ω Nodes (K)	h (mm)
(a)	0.25	0.155	16.5	9
(b)	0.30	0.150	66.0	7
(c) $\delta = 1.0$	0.33	0.33	16.5	7
(c) $\delta = 1.5$	0.33	0.33	16.5	11

observations made in the literature [34,44,45], according to which, for higher Re_W , a more round duct is expected due to the higher inertial forces of the fluid. In Fig. 9b, the evolution of both the objective function value and the area inequality constraint for $Re_W = 2$ is presented. For the optimized designs: $J = 8.86 \cdot 10^{-3} \frac{W}{m}$ for $Re_W = 2$, $J = 9.39 \cdot 10^{-4} \frac{W}{m}$ for $Re_W = 20$, and $J = 1.11 \cdot 10^{-4} \frac{W}{m}$ for $Re_W = 200$, while g_v ($\approx O(10^{-5})$) is satisfied.

The optimized design of CCTopO for $Re_W = 200$ is compared with the outcome of SPTopO. The optimized solution exhibited a binary porosity distribution, shown in Fig. 10a. To illustrate the importance of imposing accurate boundary conditions along the solid walls, the SPTopO solution was re-evaluated. The generated cut-cell mesh in Ω_f had practically the same area as the SPTopO solution ($\Delta\Omega_f \leq 10^{-3} \Omega_f^{SPTopO}$). The obtained J values are $J^{SPTopO} = 1.6 \cdot 10^{-4} \frac{W}{m}$ and $J^{r-SP} = 1.2 \cdot 10^{-4} \frac{W}{m}$ which signifies that the objective function

computed by SPTopO may have differences from that computed by generating a body-fitted mesh and re-evaluating the obtained design on it.

Comparison of J^{r-SP} and that obtained by CCTopO ($J^{CCTopO} = 1.11 \cdot 10^{-4} \frac{W}{m}$), shows that the design optimized by CCTopO exhibits 8% extra reduction in J . The solid boundaries of the CCTopO design, Fig. 10b, corresponds to a slightly shorter and wider duct and this explains the difference in the value of J .

In Figs. 11a and 11b, the sensitivity derivatives, w.r.t. the porosity field α_b in Fig. 11a and w.r.t. the material field α in Fig. 11b, are shown in the last optimization cycle for the SPTopO (Fig. 10a) and the CCTopO (Fig. 10b). The main difference between the two methods becomes clear. In SPTopO, the sensitivity derivatives are computed throughout Ω_D , whereas in CCTopO, a sensitivity map is obtained along the solid walls which is then transformed to nodal sensitivities of the material field. Thus, the sensitivity derivatives exist only in the vicinity of the current solid-fluid interface as seen in Fig. 11b and depend on the smoothing operator Eq. (3).

Furthermore, the design variables in the two methods influence the objective function differently and, therefore, an immediate comparison of scale should not be performed.

4.1.2. Single inlet-two outlet case

Following [13], this TopO was performed at $Re_W = 200$. As in the case of Section 4.1.1, a CCTopO was first performed. The so-optimized design gives $J^{CCTopO} = 0.109 \frac{W}{m}$ and a negative value of $g_v^{CCTopO} \approx O(10^{-6})$. In Fig. 12a, the evolution of the objective and area inequality constraint function of CCTopO is presented. In the first optimization

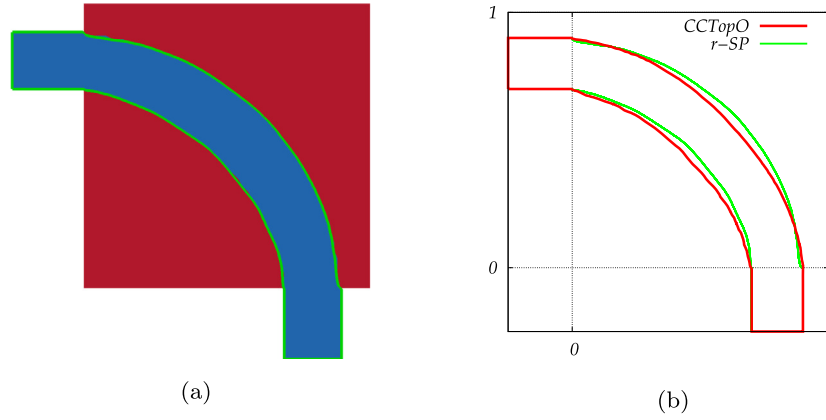


Fig. 10. Single Inlet-Single Outlet, $Re_W = 200$: (a) r-SP solid boundaries (green line) with the area with $a_b = 1$ in red and that with $a_b = 0$ in blue. (b) Comparison of the CCTopO and r-SP solid boundaries.

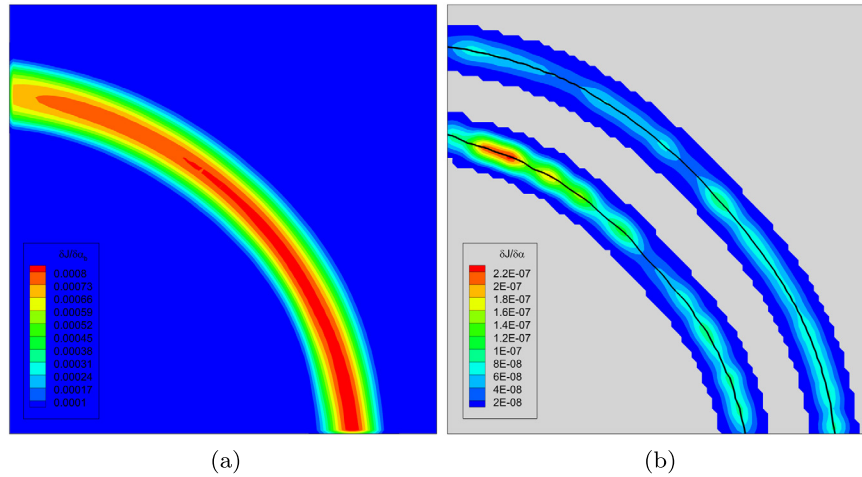


Fig. 11. Single Inlet-Single Outlet, $Re_W = 200$: Sensitivity field, w.r.t. the porosity field α_b in (a) and w.r.t. the material field α in (b), computed in the last optimization cycle using (a) SPTopO and (b) CCTopO. Derivatives at the gray areas of (b) are not computed.

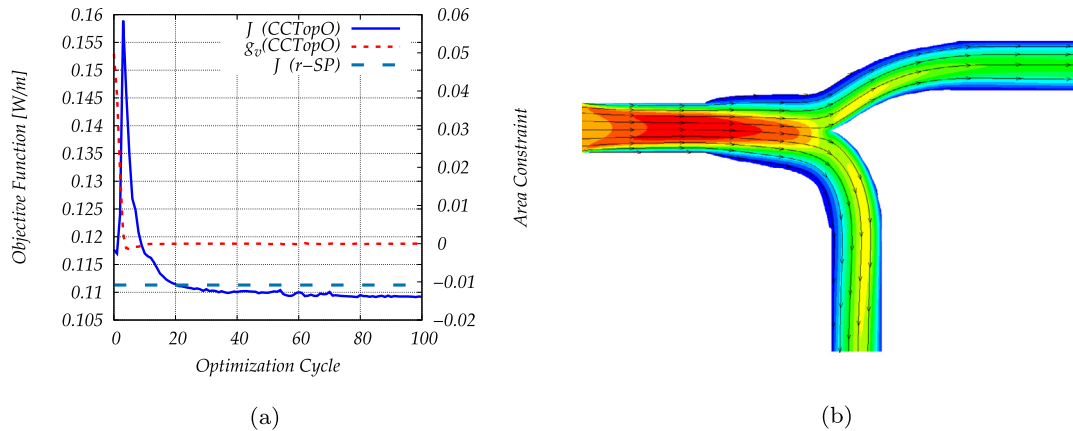


Fig. 12. Single Inlet-Two Outlet: (a) Evolution of the objective and constraint functions converging to $J^{CCTopO} = 0.109 \frac{W}{m}$ and a negative $g_v^{CCTopO} \approx O(10^{-6})$ value. (b) Velocity magnitude $\|u\| \in [0, 1.2]$ iso-areas for the optimal design of CCTopO with $d^r = 5$.

cycles, designs failed to meet the area constraint. Focusing in obtaining a feasible design, the fluid domain was restricted leading to an increase in the J values. As soon as a feasible design was found, the reduction in J starts. At the 40th optimization cycle, the optimized design shape was mostly settled, with slight changes occurring at the area where the duct splits. Due to the higher sensitivities in that area, minor changes

in the design variables alter the splitting angle affecting the obtained J values.

Also, a SPTopO was conducted and this resulted to a porosity distribution in Ω_D that was used to trace the solid walls in order to re-evaluate this solution using the cut-cell method. Again, the r-SP fluid area was the same as that of SPTopO ($\Delta\Omega_f \approx 10^{-3} \Omega_f^{SPTopO}$). The com-

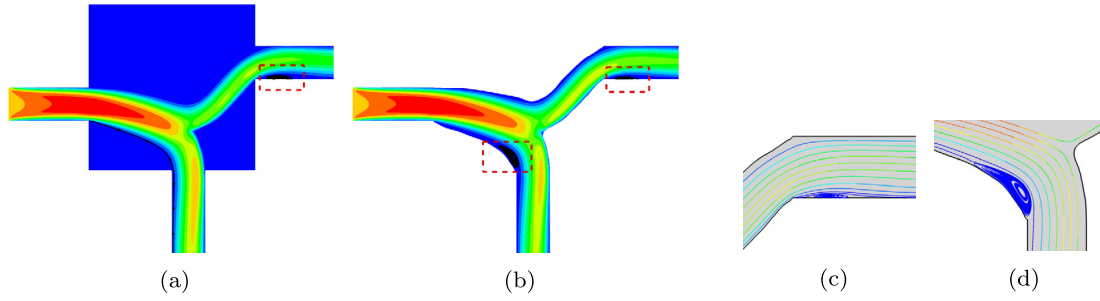


Fig. 13. Single Inlet-Two Outlet: Velocity magnitude $\|\mathbf{u}\| \in [0, 1.2]$ iso-areas computed by the (a) SPTopO and (b) r-SP. (c) Flow separation close to the right outlet. (d) Second flow separation zone appearing only in r-SP solution.

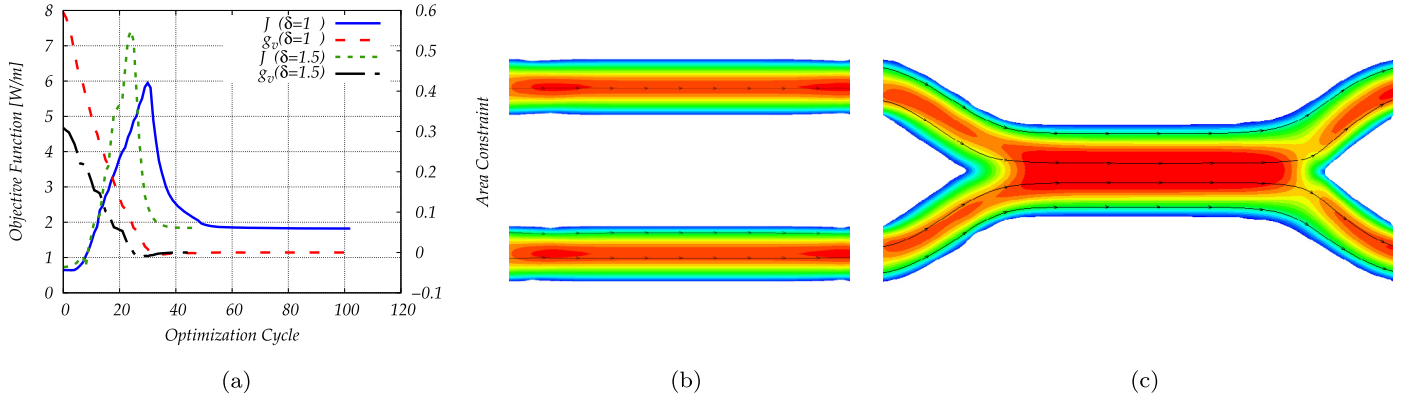


Fig. 14. Two Inlet-Two Outlet: (a) Evolution of the objective and constraint functions for $\delta = 1.0$ and $\delta = 1.5$. Velocity magnitude $\|\mathbf{u}\| \in [0, 1.5]$ iso-areas in the optimized ducts for (b) $\delta = 1.0$ and (c) $\delta = 1.5$. Inlet/outlet extensions are not shown.

parison of velocity magnitude iso-areas for both the SPTopO and r-SP, Figs. 13a and 13b, leads to a couple of interesting observations. First, a separation zone manifests in both solutions near the right outlet due to the steepness of the connecting duct, Fig. 13c. Furthermore, in the r-SP solution, a second separation zone, non-existent in the SPTopO solution, can be seen in Fig. 13d due to the accurate boundary conditions imposed along the solid walls. This difference in the flow patterns reflects upon the difference (by approximately 3%) in J , since $J^{SPTopO} = 0.115 \frac{W}{m}$ and $J^{r-SP} = 0.112 \frac{W}{m}$.

Comparing the J values of r-SP and CCTopO, Fig. 12a, an improvement (by 2%) in J is observed but it is less important than in Case 4.1.1. However, the attained designs, shown in Figs. 12b and 13b, are quite different. In the shape obtained by r-SP, the first part of the duct is inclined towards the bottom outlet, giving rise to the steepened duct that connects to the right outlet and causes flow separation. In contrast, the CCTopO retains an almost horizontal first part that splits into a wider lower and a narrower upper duct. The part connecting to the right outlet changes gradually, avoiding thus flow separation. Furthermore, the first part splits into two parts much earlier and is also wider near the inlet, decelerating the bulk flow. Overall, the optimal design of the CCTopO is more inline with the optimized design presented in [13], using ShpO, on a much denser body-fitted mesh, after transitioning from a TopO to ShpO. This highlights the ability of the proposed method to solve TopO problems while retaining the accuracy of CFD simulations that impose accurate boundary conditions, as it is usually the case in ShpO.

4.1.3. Two inlet-two outlet case

The two inlet-two outlet TopO case was optimized for two different δ values, Fig. 8c, both at $Re_W = 17$. This problem has been studied in [3,5,45] and, depending on δ , different fluid path layouts emerge, demonstrating TopO's flexibility to compute topologically different de-

signs. For $\delta = 1.0$, the two flow paths remain separate, while for a longer distance, $\delta = 1.5$, the two flow paths merge into a wider duct.

In Fig. 14a, the evolution of the objective and constraint functions are shown for both cases. The stopping criterion was reached after 103 ($\delta = 1.0$) and 47 ($\delta = 1.5$) cycles. A larger number of optimization cycles was required for $\delta = 1.0$ since the solid boundaries between the inlets needed to merge with those between the outlets to form two separate fluid paths [5]. In Figs. 14b and 14c, the optimized designs are shown along with the velocity magnitude iso-areas. The obtained designs match those of [3,5,45]. The objective function values computed by the CCTopO method were $J^{\delta=1.0} = 1.82 \frac{W}{m}$ and $J^{\delta=1.5} = 1.84 \frac{W}{m}$.

4.2. 3D manifold TopO

In this section, the CCTopO method and software is used for the design of a 3D manifold. The case consists of a single inlet at the top and four symmetrically placed outlets on the $z = 0$ plane, Fig. 15. The Reynolds number, based on the inlet diameter D , is $Re_D = 4$. A fixed inlet velocity $\mathbf{u}_{in}$ and a uniform outlet reference pressure p_{ref} were specified. The problem was setup with a target $\frac{\Omega_f^*}{\Omega_D} = 0.2$ corresponding to $V_{tar} = 0.05$ and an initially fluid Ω_D domain. The Ω mesh consisted of approximately 2.2M nodes that lead to $h = 0.015$ and an initial Ω_f mesh with 250K cells.

The optimization was ended after 29 cycles with $J = 3.49 \cdot 10^{-3} W$ and a negative g_v ($\approx O(10^{-6})$) obtaining the design shown in Fig. 16. From the evolution of J and g_v , Fig. 17, it can be seen that the initial solution corresponds to an infeasible design with quite a low J value, as Ω_D (Fig. 16) was initialized as fluid. In order to satisfy the volume constraint, the fluid domain was restricted, accelerating flow near the inlet and outlets which, in turn, increased wall shear stresses. This also explains the increase in the J value up to the appearance of a feasible design. Afterwards, GCMMA focused on minimizing J by computing

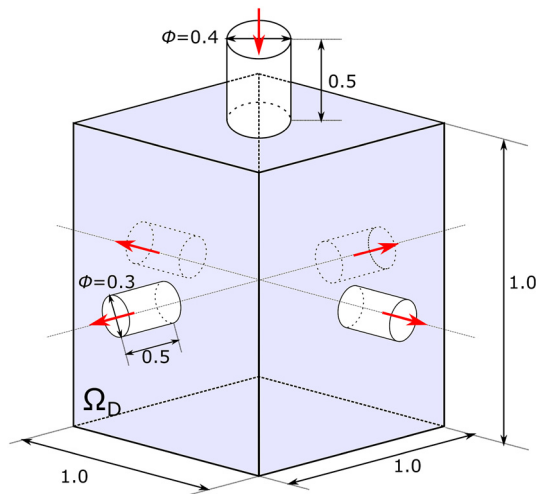


Fig. 15. 3D manifold: Problem description for the design of a single inlet-multiple outlet TopO case. The α field is allowed to change in the highlighted cubic domain, Ω_D . All dimensions are in m .

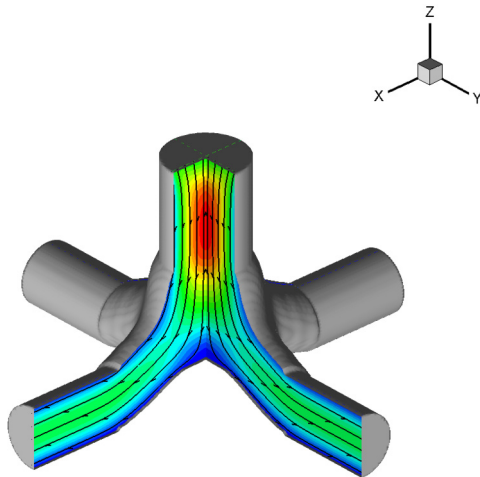


Fig. 16. 3D manifold: Velocity magnitude $\|\mathbf{u}\| \in [0, 0.19]$ iso-areas along the $x = 0.5$, $y = 0.5$ planes.

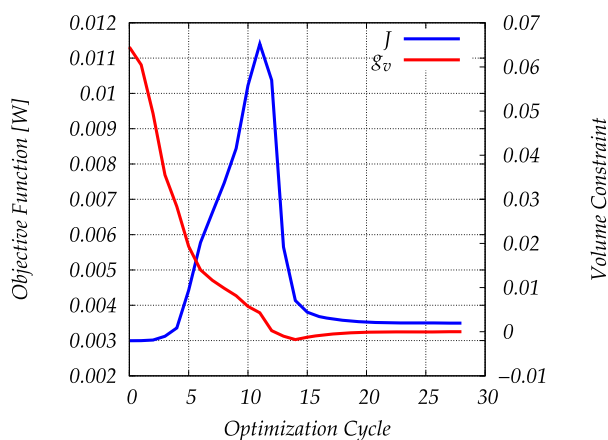


Fig. 17. 3D manifold: Evolution of the objective and constraint functions using the CCTopO method.

better suited feasible designs. This was achieved by further restricting the corner areas to expand the previously restricted cross sections near the inlet and outlets. In other words, this process eliminated stagnated flow areas by allocating their “effective” volume near the outlet positions, resulting in the obtained tuboid design. In the optimized design, the flow is attached and equally distributed to the four outlets as their volumetric flow rates were practically the same.

Fig. 16 shows the velocity magnitude iso-areas of the optimized design along the $x = 0.5$ and $y = 0.5$ planes indicating the absence of flow separation. Figs. 18a and 18b present xy and yz planar views of the optimized design showing the resulting connections. In Fig. 18c, the solid boundary at the bottom side, with a close up view of some solid faces, is shown. Each triangle corresponds to a solid face on which no-slip conditions were imposed.

5. Conclusions

A cut-cell based topology optimization method that reconstructs the sought solid boundaries during the optimization process in order to impose accurate boundary conditions was presented. A background material field determines the design variables of the problem. Fluid-solid interfaces are traced by processing the material field on a Cartesian mesh and this process results to the definition of cut-cells. During the optimization, the continuous adjoint method is used to compute a sensitivity map over the aforementioned interfaces, while the implicit geometry description allows for topological changes to occur. The gradient of the objective function w.r.t. the material field is extracted using the chain rule on the obtained sensitivity map.

The proposed method was used for the optimization of three 2D benchmark TopO cases and a 3D manifold, all of them with the same objective function. In two cases, the optimized designs were compared with those obtained by a standard porosity-based TopO. For a fair comparison, optimized designs using the porosity-based TopO were re-evaluated on the cut-cell flow solver using the mapping procedure of the proposed method. It was made clear that there is a need to re-evaluate any optimized by porosity-based TopO design. Based on these comparisons, the cut-cell based TopO was shown to provide better-performing designs with reduced objective function values, by approximately 8% in the first and 2% in the second case, when compared with the re-evaluated ones. In the single inlet-double outlet case, the optimized designs had visible differences. The design obtained by the proposed method was in-line with the results of [13], in which a shape optimization was performed starting from the topologically optimized design.

The solid boundary reconstruction using the cut-cell method will allow for design criteria, defined specifically on solid walls, or the inclusion of turbulence, where near-wall effects are of essence, to be better considered. The crisp solid-fluid interface allows extension of the TopO to adjoint optimization in aero-structural, conjugate heat transfer problems or any other problem with two disciplines strongly interacting along their interface.

Extension of the proposed CCTopO method to other objective functions is straightforward, provided that the corresponding adjoint formulation is available. Of course, the proposed method can also be applied with discrete adjoint.

Acknowledgement

This research is co-financed by Greece and the European Union (European Social Fund-ESF) through the Operational Programme “Human Resources Development, Education and Lifelong Learning” in the context of the project “Strengthening Human Resources Research Potential via Doctorate Research” (MIS-5000432), implemented by the State Scholarships Foundation (IKY).

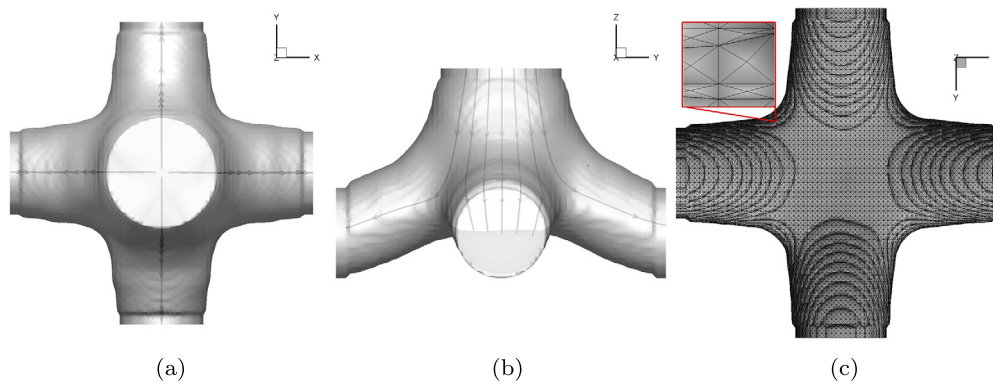


Fig. 18. 3D manifold: View normal to the (a) xy -plane and (b) yz -plane for the optimized design with selected velocity streamlines, showing the symmetrical solution. (c) Bottom view of the xy -plane with a close-up on some solid faces of the generated cut-cell mesh during the last optimization cycle.

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